



UNIVERSITY of
RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF PHYSICS/SCHOOL OF SCIENCE

TRIMESTER ONE EXAM
ACADEMIC YEAR 2023-2024

DEPARTMENT: PHYSICS

YEAR LEVEL OF STUDY: FIRST YEAR

STUDY GROUP: CS, IT, EPE, CE, CI, TE, ETE, SGE, WREE, EE, IS

Engineers at Nyarugenge campus and CS, MEE engineers at Gako campus.

MODULE CODE: PHY1163

MODULE NAME: Physics for Engineers I

MODULE WEIGHTING: 10 credits

DATE: ...

DURATION: 2 HOURS

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains 5 questions to be answered in 2 Sections: Section A and Section B.
 2. Section A is compulsory; you must answer all 2 questions.
 3. Section B allows a choice; select any 2 questions from the 3 questions given and answer them.
 4. Write both your Registration and your Examination Numbers on the Examination Booklets provided.
 5. Write all your answers only in the examination booklet provided.
 6. Do not write on this question paper.
 7. Start each question on a fresh page.
 8. No unauthorized material is allowed in the examination room.
 9. You may use the following information if needed: Gravity $g = 9.80\text{m/s}^2$;
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SECTION A: COMPULSORY QUESTIONS: [20 marks]

Question N° 1 [10 marks]

- a) Differentiate the fundamental quantities from derived quantities in physics and give an example for each quantity. (3 marks)
- b) What is the difference between scalar and vector physical quantities? (2 marks)
- c) An object moves along the x-axis according to the equation $x(t) = 3t^2 - 2t + 3$, where x is in meters and t is in seconds.

- Determine:
- i) the average speed between $t = 2s$ and $t = 3s$ (1 mark)
 - ii) The instantaneous speed at $t = 2s$ and $t = 3s$ (1 mark)
 - iii) The average acceleration between $t = 2s$ and $t = 3s$ (1 mark)
 - iv) The instantaneous acceleration at $t = 2s$ and $t = 3s$ (1 mark)
 - v) At what time is the object at rest? (1 mark)

Question N° 2 [10 marks]

- a) State Newton's laws of motion and their mathematical expressions. (3 marks)
- b) State the work-kinetic energy theorem and prove its mathematical expression. (2 marks)
- c) In the Physics domain
- i) Explain a rigid body (2 marks)
 - ii) What is the difference between plastic body and elastic body (3 marks)

SECTION B: YOU CAN ATTEMPT ANY FOUR QUESTIONS: [30marks]

Question N° 3 [15 marks]

- a) A golf ball is hit at an angle of 30° above horizontal with a speed of 44 m/s.
- i) How high does it rise? (3 mark)
 - ii) How long is it in the air? (2 mark)
 - iii) How far does it travel horizontally? (3 mark)
- b) The four masses shown in Figure 1 below are each 2 kg. They form a square with side lengths of 1 meter. Calculate the moment of inertia of the system of masses about a perpendicular axis located at the center of the square. (3 marks)

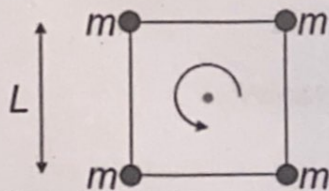


Figure 1: Point Masses in Square Formation.

- (4 marks)

a) Distinguish between fluid

- (3 marks)

(3 marks)

(1mark)

(1mark)

(1mark)

(3 marks)

, the density of water = 1000
(3 marks)

c) Distinguish the long

- (3marks)**

(3 marks)

(2 marks)

(2 marks)

(2 marks)

(3 marks)

time $t=20$ s

END OF QUESTIONS